THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. To find whether the volume of water affects its boiling time, a student boils 100 mL in a steel pot on a strong flame and 200 mL in a glass beaker on a weak flame. Why is this test unfair? 1

(a) the water is heated at all
(b) the volumes of water are different
(c) time is measured with a clock
(d) the container and the flame also differ along with the volume

2. A scientist finds a plant with parallel veins, fibrous roots and a soft stem. Which is most probable? 1

(a) a peepal tree **(b)** a grass **(c)** a rose bush **(d)** a mango tree

3. A food gives a blue-black colour with iodine and an oily patch on paper. It contains: 1

(a) fat only **(b)** starch only
(c) starch and fat **(d)** protein and fibre

4. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) all are iron
(b) all are plastic
(c) all are magnets
(d) P and R are magnets, Q is magnetic material

5. A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

(a) 320 cm **(b)** 100 cm **(c)** 160 cm **(d)** 200 cm

6. A student heats the ends of four rods P, Q, R, S, each with a wax-fixed pin at the same distance. The pins fall at 10 s, 25 s, 40 s and 60 s. The best conductor is: 1

(a) R **(b)** S **(c)** P **(d)** Q

7. Thermometer A reads 0 °C in melting ice and 100 °C in boiling water. Thermometer B reads 2 °C and 98 °C. For the same hot water at actual 50 °C, B would most likely read: 1

(a) 48 °C **(b)** 52 °C **(c)** 50 °C **(d)** 54 °C

8. When ice at 0 °C is heated, the temperature stays at 0 °C till all ice melts. This shows that: 1

(a) the thermometer is faulty **(b)** ice cannot be heated
(c) the heater is weak **(d)** heat is used to change the state

9. A mixture of iron filings, common salt, and fine sand is given. Which sequence is the most efficient? 1

(a) add water, evaporate, magnet
(b) filter, magnet, sieve
(c) magnet, add water and filter, evaporate filtrate
(d) evaporate first, filter, magnet

10. A candle burns, melts, gives out waste gases and its flame responds to wind. Why is it still non-living? 1

(a) it does not grow or reproduce on its own **(b)** it melts
(c) it is hot **(d)** it makes light

11. A student collected 10 mL of gas from a jar over water after a candle burnt out. The candle extinguished because: 1

(a) the jar broke **(b)** all nitrogen was used
(c) no gas remained **(d)** most oxygen was used

12. If sunlight takes 8 minutes to reach the Earth, the Sun's distance in terms of light travel time is: 1

(a) 8 light-years **(b)** 8 light-hours **(c)** 8 light-seconds **(d)** 8 light-minutes

13. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather **(b)** sunburn causes ice cream sales
(c) it is magic **(d)** ice cream causes sunburn

14. Which of the following is a correct inference about a tiger's eyes in front of the face? 1

(a) helps fly **(b)** helps see all sides
(c) helps judge distance for hunting **(d)** helps to swim

15. Why do nutritionists advise limiting packaged drinks? 1

(a) they contain much sugar and few nutrients
(b) they have vitamins
(c) they have fibre
(d) they contain protein

16. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not **(b)** both are shiny
(c) both are metals **(d)** both rust

17. In a pendulum experiment, 20 oscillations took 36 s. A second student gets 18 s for 10 oscillations. Both agree because: 1

(a) time period is 1.8 s in both **(b)** he is wrong
(c) time period differs **(d)** 36 s vs 18 s differ

18. A boy tests four materials A, B, C, D with a torch. Light passes fully through A, partly through B, not at all through C and D. Which can be used for a window to see out and still keep privacy? 1

(a) C **(b)** A **(c)** B **(d)** D

19. Rod A conducts heat twice as fast as rod B. If wax pins fall from A in 20 s from the heated end, the same pin on B would fall in about: 1

(a) 40 s **(b)** 20 s **(c)** 5 s **(d)** 10 s

20. Water of 100 g at 0 °C is converted to ice at 0 °C. The mass of ice formed is: 1

(a) more than 100 g **(b)** 100 g **(c)** 0 **(d)** less than 100 g

21. 100 g of a mixture contains 20 g of iron, 30 g of sand and 50 g of salt. After separation, the mass of sand obtained is expected to be: 1

(a) 50 g **(b)** 100 g **(c)** 20 g **(d)** 30 g

22. Out of a tree, a bird and a pebble, which one is non-living? 1

(a) pebble **(b)** bird
(c) tree **(d)** both bird and tree

23. Which of these actions improves soil fertility naturally? 1

(a) adding plastic **(b)** adding compost **(c)** removing worms **(d)** burning plastics

24. Mass of an object is 60 kg on Earth. On the Moon (gravity one-sixth), its mass is: 1

(a) 360 kg **(b)** 6 kg **(c)** 10 kg **(d)** 60 kg

25. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) toss a coin **(b)** compare the methods and repeat carefully
(c) ignore both **(d)** believe one group

26. In a forest, if all insects vanish, what is most likely to happen? 1

(a) animals will grow bigger **(b)** forest will grow faster
(c) many plants will fail to pollinate **(d)** all plants will grow faster

27. A paper rubbed with a food sample shows a translucent oily patch that stays even after drying. The food contains: 1

(a) fat **(b)** starch **(c)** fibre **(d)** sugar

28. To make a magnet from an iron bar by the single-touch method, the bar should be stroked: 1

(a) by hitting it with the magnet
(b) back and forth with both poles
(c) in a circle with the magnet
(d) always in the same direction with the same pole of a magnet

29. A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

(a) 12.5 cm **(b)** 14.5 cm **(c)** 2.0 cm **(d)** 10.5 cm

30. Why does a heavy ship made of steel float? 1

(a) steel is lighter than water **(b)** ships have no weight
(c) its shape displaces much water **(d)** water is sticky

31. The best conductor-insulator pair for a cooking pot and its handle is: 1

(a) glass and iron **(b)** plastic and aluminium
(c) aluminium and plastic **(d)** wood and plastic

32. A student claims that the water in a cup disappeared. The correct statement is: 1

(a) it changed to vapour and mixed with air **(b)** it became oxygen
(c) it vanished **(d)** it was destroyed

33. Why is alum added during water purification before filtration? 1

(a) it kills all germs **(b)** it adds taste
(c) it dissolves salt **(d)** it makes tiny particles clump and settle

34. A dry seed lies unchanged in a packet for months but sprouts after being watered. This shows that: 1

(a) water makes seeds alive from stone
(b) a living thing can stay inactive until conditions suit it
(c) seeds are non-living
(d) sprouting is random

35. A town has 1,00,000 vehicles each emitting smoke. Which step reduces the problem most? 1

(a) more public transport **(b)** less trees
(c) more cars **(d)** more roads only

36. Which of the following explains day and night on Earth? 1

(a) Earth rotates on its axis **(b)** the Sun moves around the Earth
(c) Earth revolves around the Sun once a day **(d)** the Moon blocks the Sun daily

37. In an experiment on how salt affects the boiling point, two things were changed: salt amount and heater power. The error is: 1

(a) more than one variable was changed **(b)** too many readings
(c) salt is a solid **(d)** water is cold

38. An animal with webbed feet, lungs and moist skin that lives near ponds and jumps on land is most likely a: 1

(a) lion **(b)** crow **(c)** frog **(d)** camel

39. A student tested three foods P, Q, R with iodine. Only Q turned blue-black. Which statement is correct? 1

(a) All three contain starch **(b)** P and R contain starch
(c) Q contains fat **(d)** Q contains starch

40. A freely hung bar magnet has one end painted red; the red end settles pointing towards the geographic south. The red end is the: 1

(a) south-seeking pole **(b)** north-seeking pole
(c) neutral centre **(d)** non-magnetic part

41. **Assertion (A):** A hypothesis is a guess that can be tested. 1
Reason (R): An experiment can show whether a hypothesis is supported or not.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Wheat plants have fibrous roots. 1
Reason (R): Plants with fibrous roots generally have leaves with parallel venation.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Proteins are called body-building nutrients. 1
Reason (R): Proteins help in growth and repair of the body.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** A freely suspended bar magnet always comes to rest along the north-south direction. 1
Reason (R): The Earth behaves like a huge magnet.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The SI unit of length is the metre. 1
Reason (R): 1 metre is equal to 100 centimetres.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** Clear glass is a transparent material. 1
Reason (R): Light passes through transparent materials almost completely.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The sense of touch is not a reliable way to judge temperature. 1
Reason (R): Our skin can be misled by what we touched before, so the same water may feel hot or cold.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Evaporation takes place at all temperatures. 1
Reason (R): Boiling takes place only at a fixed temperature, about 100 degrees C.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Winnowing is used to separate heavier grains from lighter husk. 1
Reason (R): Wind blows away the lighter husk while the heavier grains fall down.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** A stone is a non-living thing. 1
Reason (R): A stone does not grow, breathe or respond to changes around it.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**